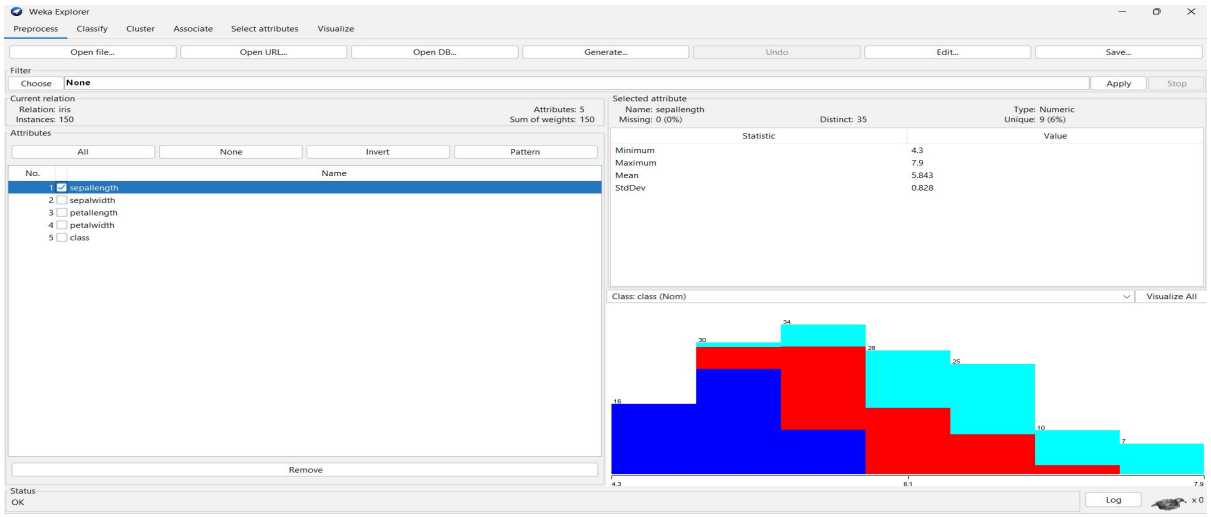
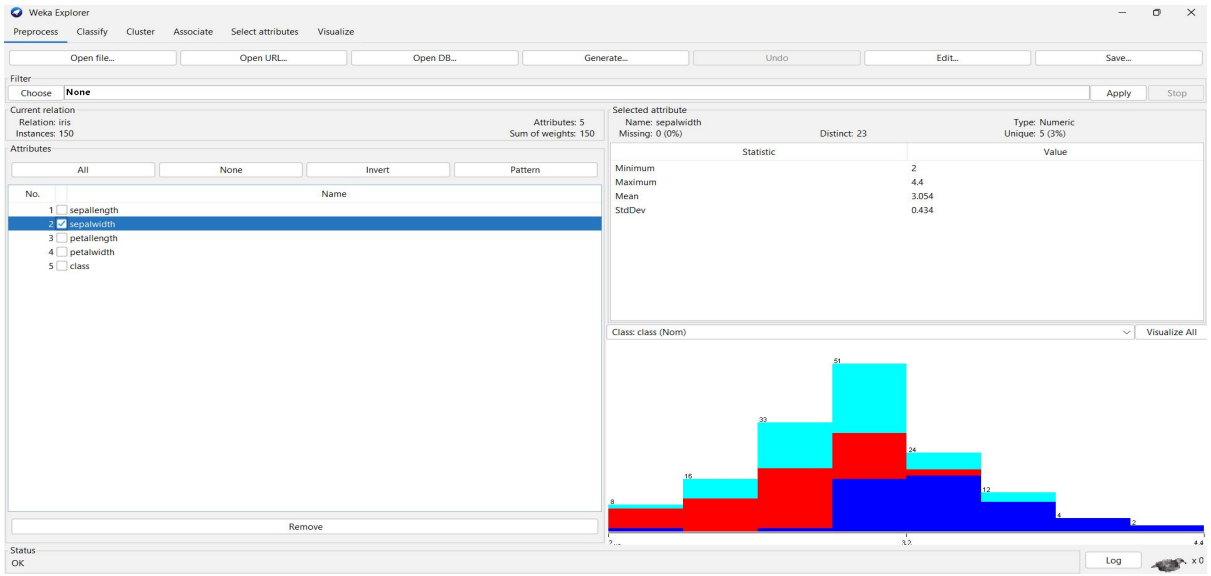
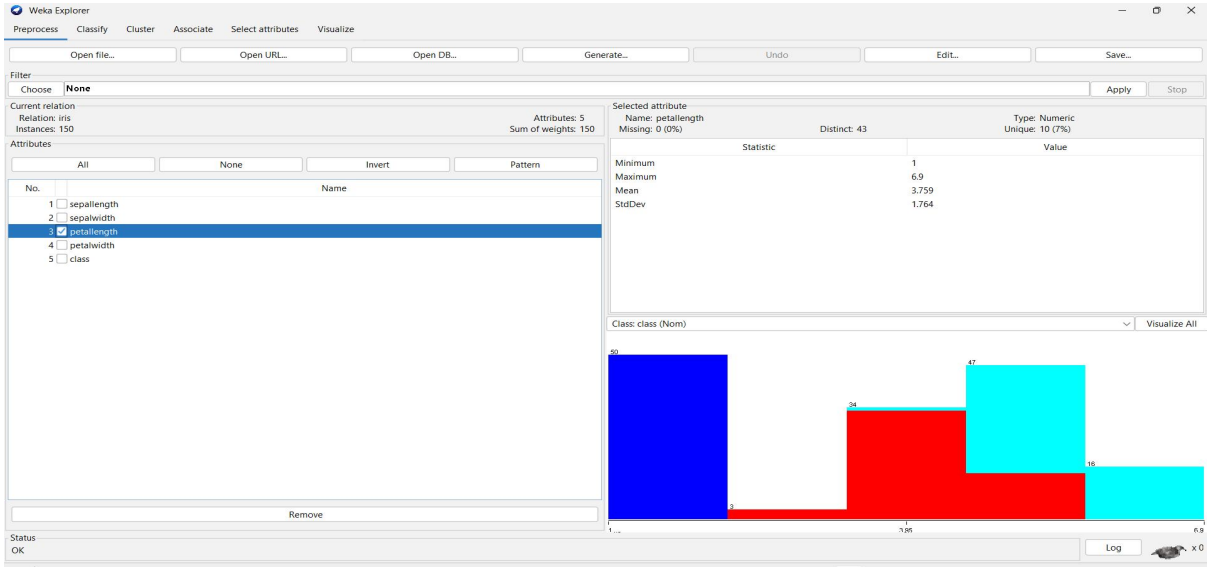
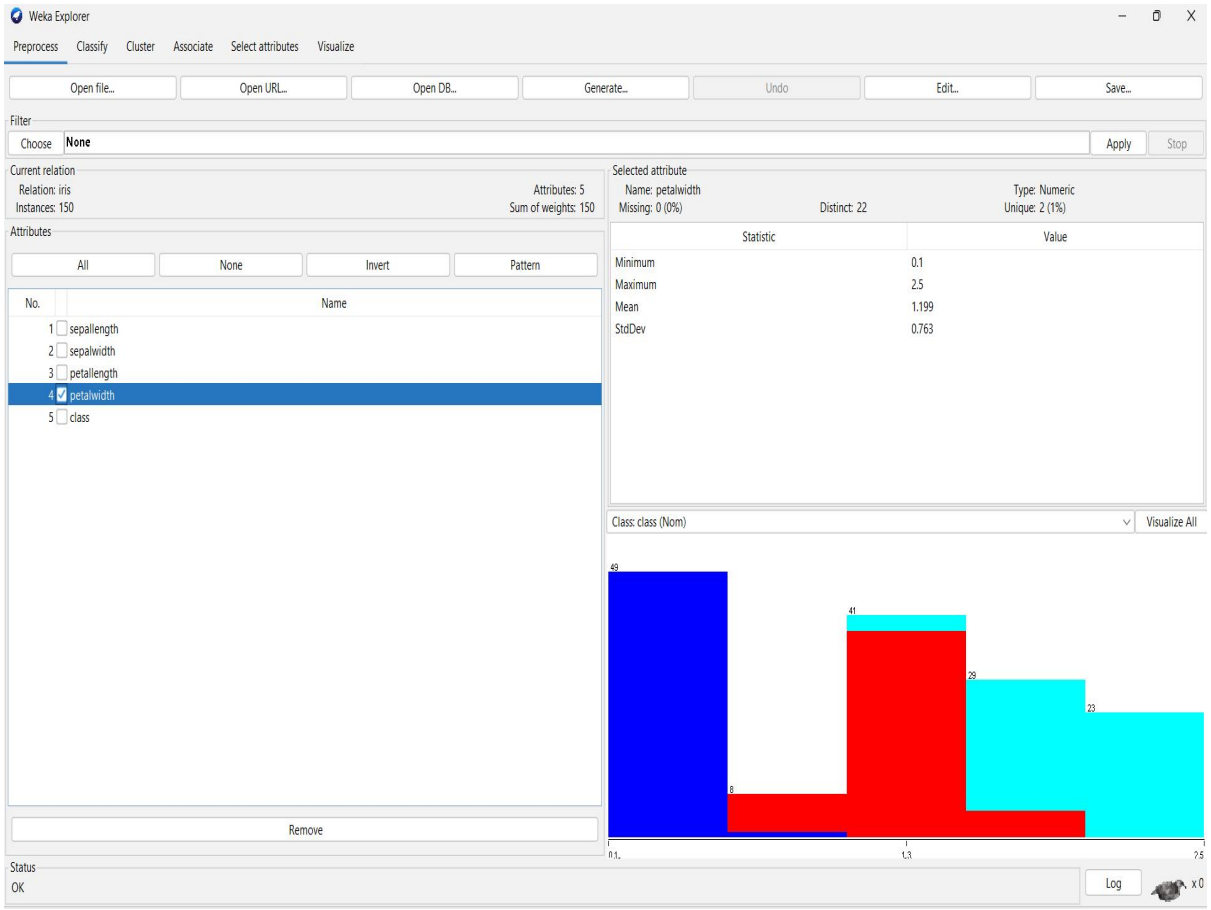
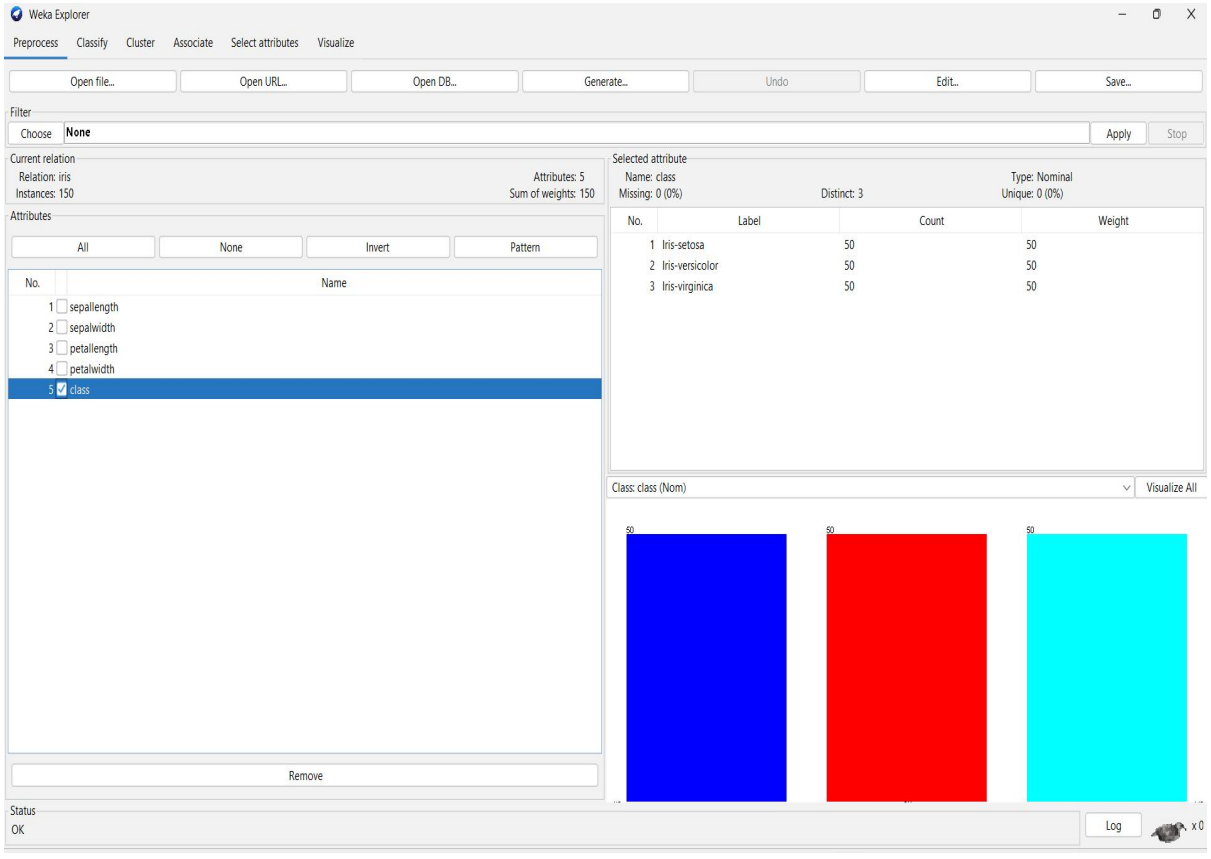
EXPERIMENT 21 : DATA PREPROCESSING AND ANALYSIS FOR DATASET USING WEKA



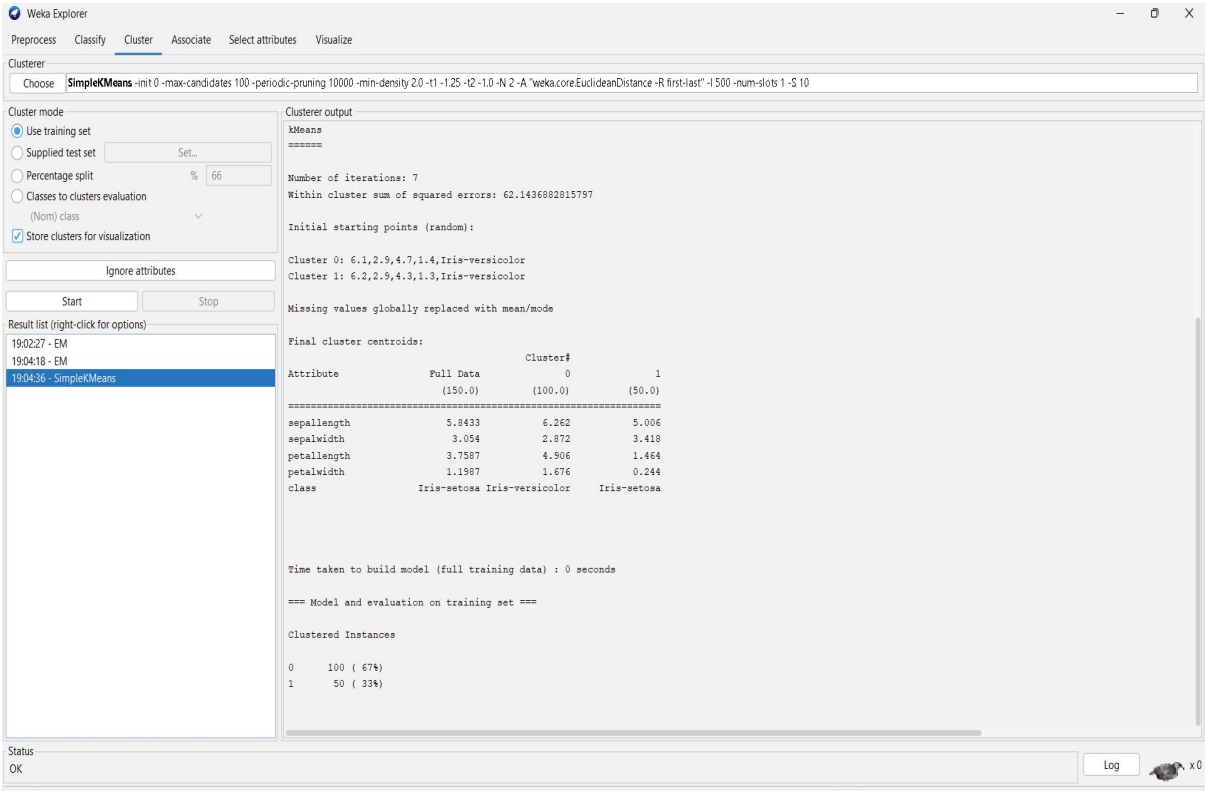


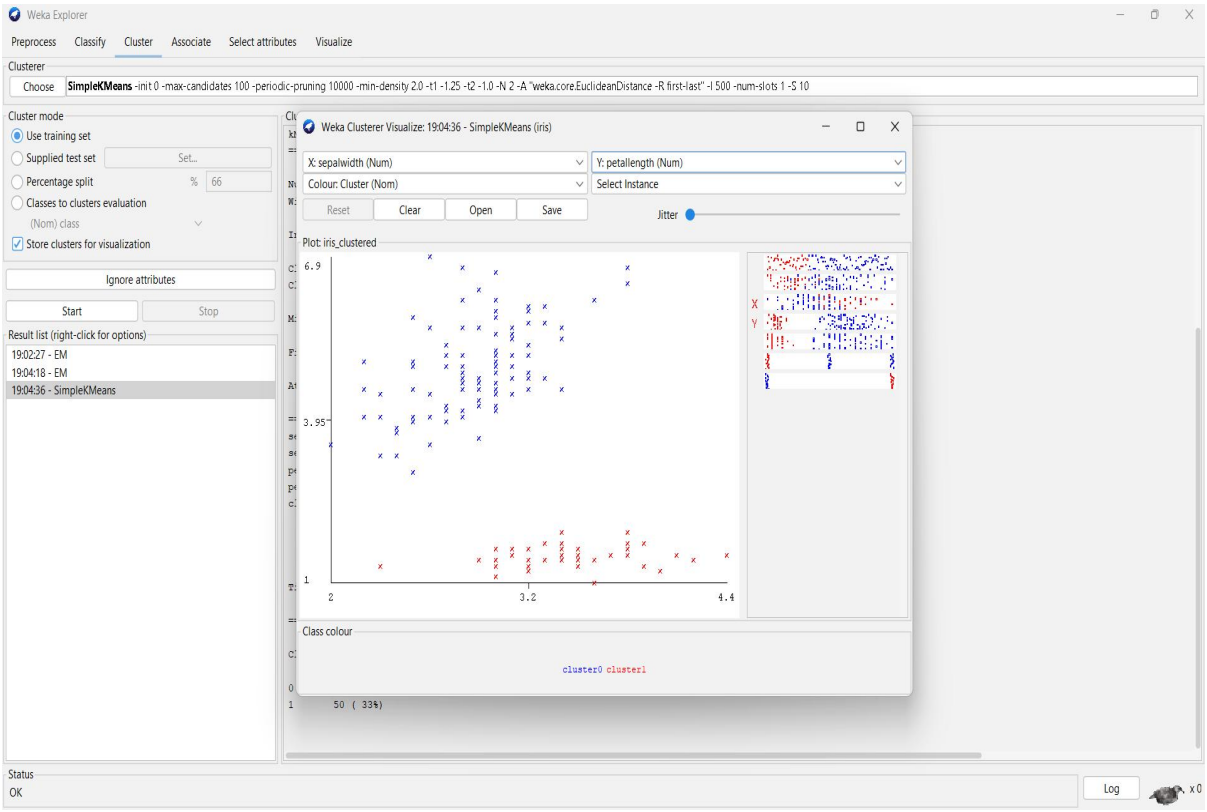



Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Open file...
Open URL...
Open DB...
Generate...
Undo
Edit...
Save...
Filter
Choose
None
Apply
Stop
Current relation
Relation: iris
Instances: 150
Attributes: 5
Sum of weights: 150
Attributes
All
None
Invert
Pattern
No.
Name
1 sepallength
2 sepalwidth
3 petallength
4 petalwidth
5 class
Remove
Selected attribute
Name: petalwidth
Missing: 0 (0%)
Distinct: 22
Type: Numeric
Unique: 2 (1%)
Statistic
Value
Minimum 0.1
Maximum 2.5
Mean 1.199
StdDev 0.763
Class: class (Nom)
Visualize All
49
8
41
29
23
0.1
1.3
2.5
Status
OK
Log
x 0

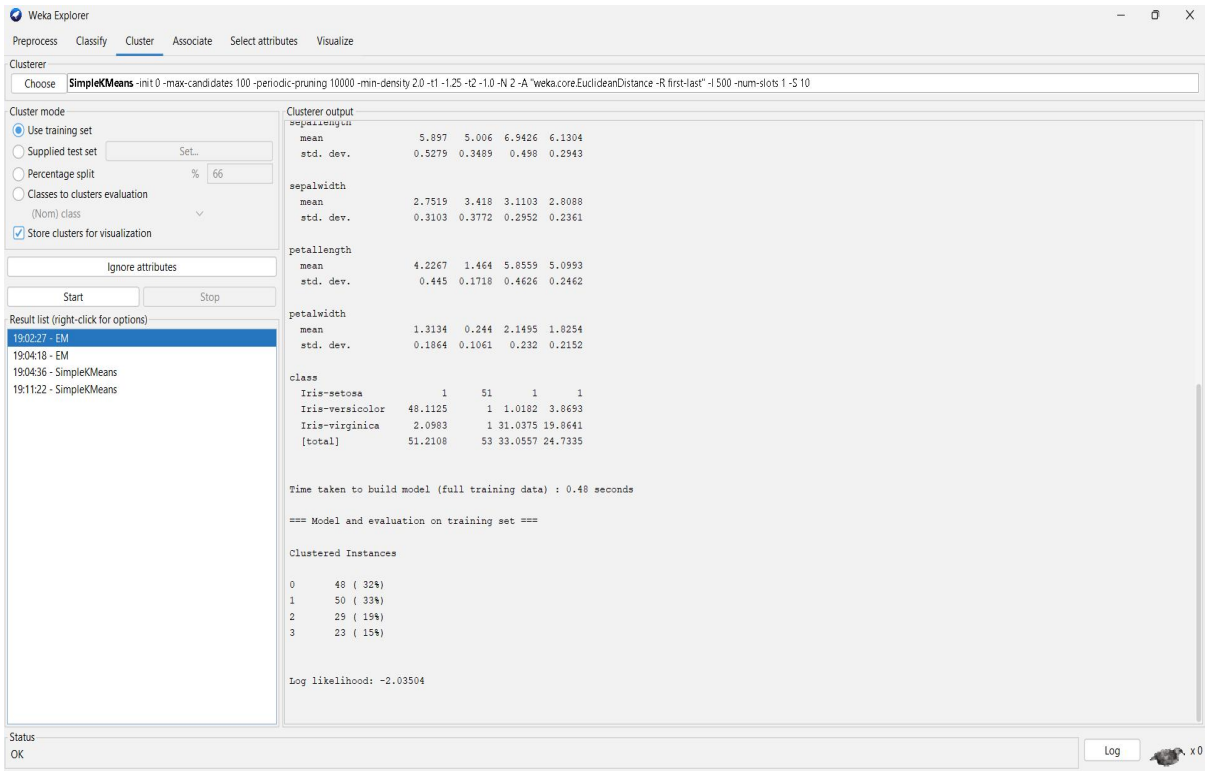

Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Open file...
Open URL...
Open DB...
Generate...
Undo
Edit...
Save...
Filter
Choose
None
Apply
Stop
Current relation
Relation: iris
Instances: 150
Attributes: 5
Sum of weights: 150
Attributes
All
None
Invert
Pattern
No.
Name
1 sepallength
2 sepalwidth
3 petallength
4 petalwidth
5 class
Remove
Selected attribute
Name: class
Missing: 0 (0%)
Distinct: 3
Type: Nominal
Unique: 0 (0%)
No.
Label
Count
Weight
1 Iris-setosa 50 50
2 Iris-versicolor 50 50
3 Iris-virginica 50 50
Class: class (Nom)
Visualize All
50
50
50
Status
OK
Log
x 0

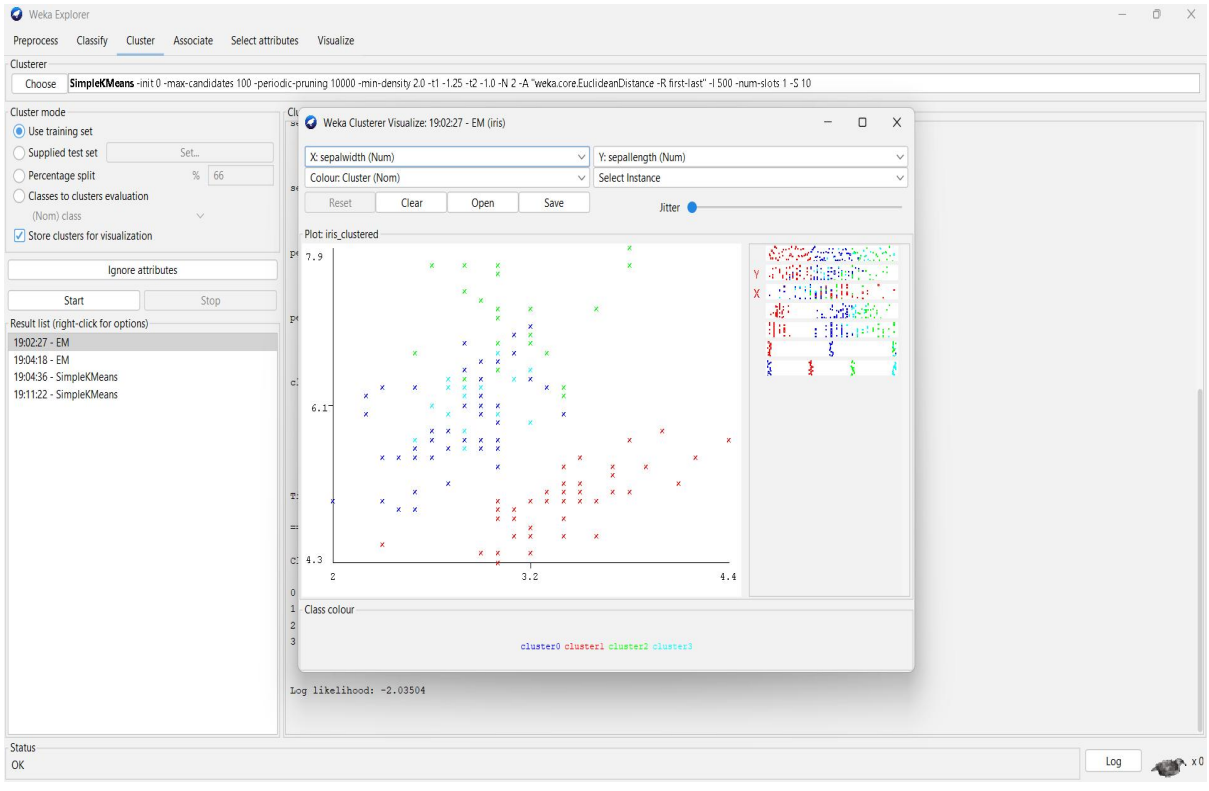
EXPERIMENT 22 : DATA SEGMENTATION BY K-MEANS CLUSTER USING WEKA AND R-TOOL



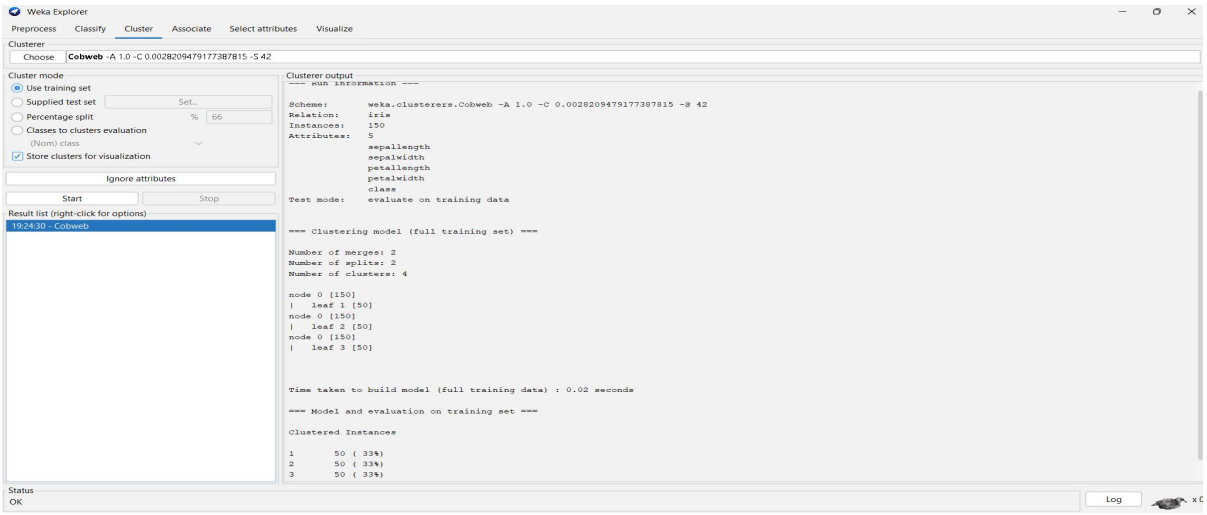


EXPERIMENT 23 : DATA SEGMENTATION BY EXPECTATION MAXIMISATION ALGORITHM THROUGH WEKA

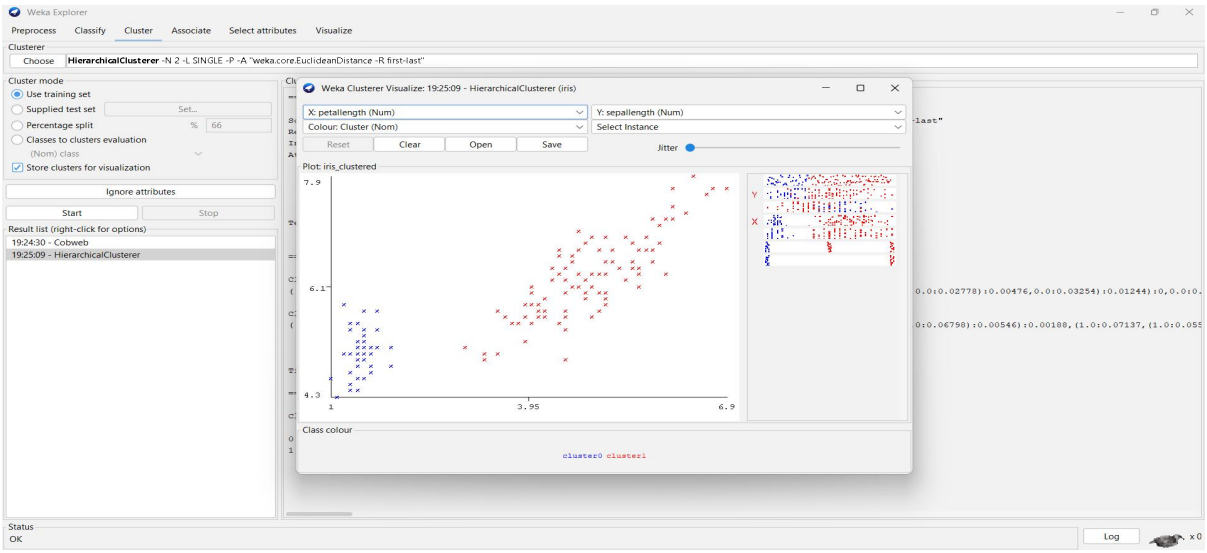




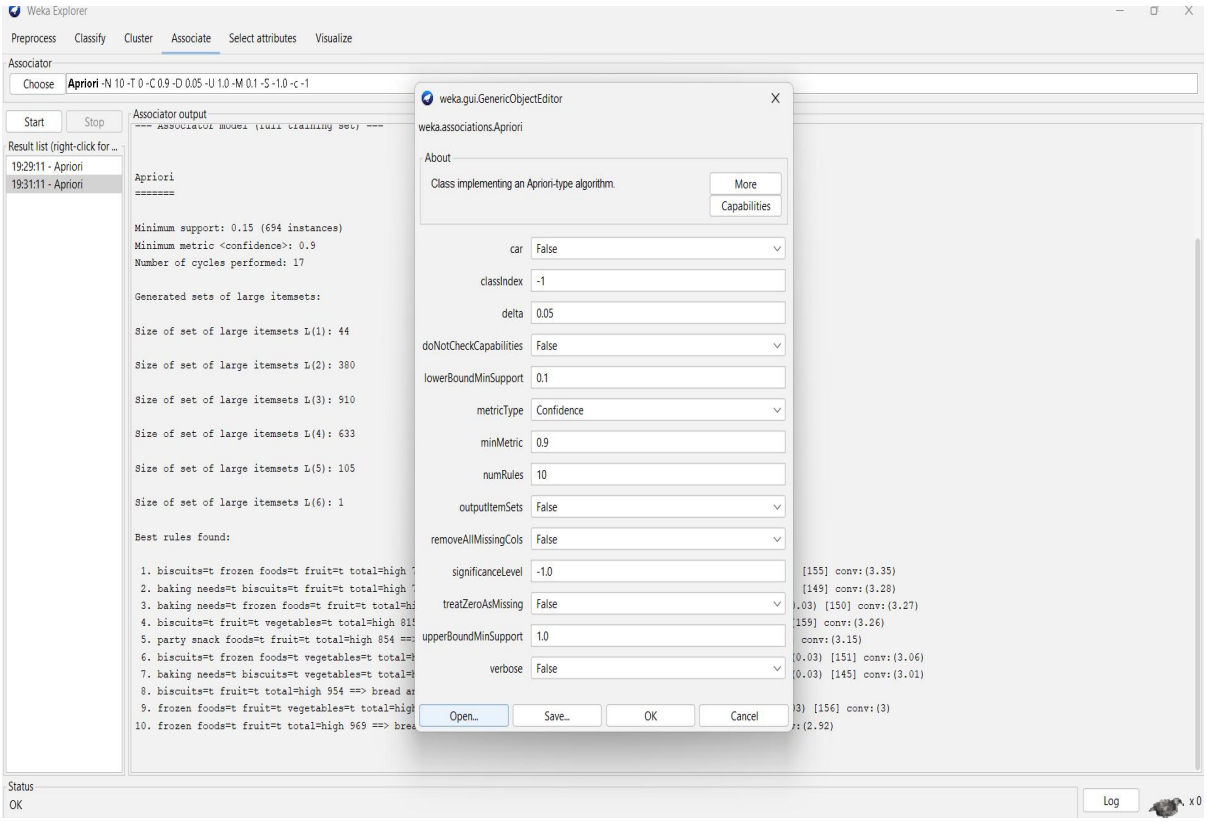
EXPERIMENT 24 : DATA SEGMENTATION BY COBWEB – HIERARCHIAL CLUSTERING ALGORITHM USING WEKA TOOL

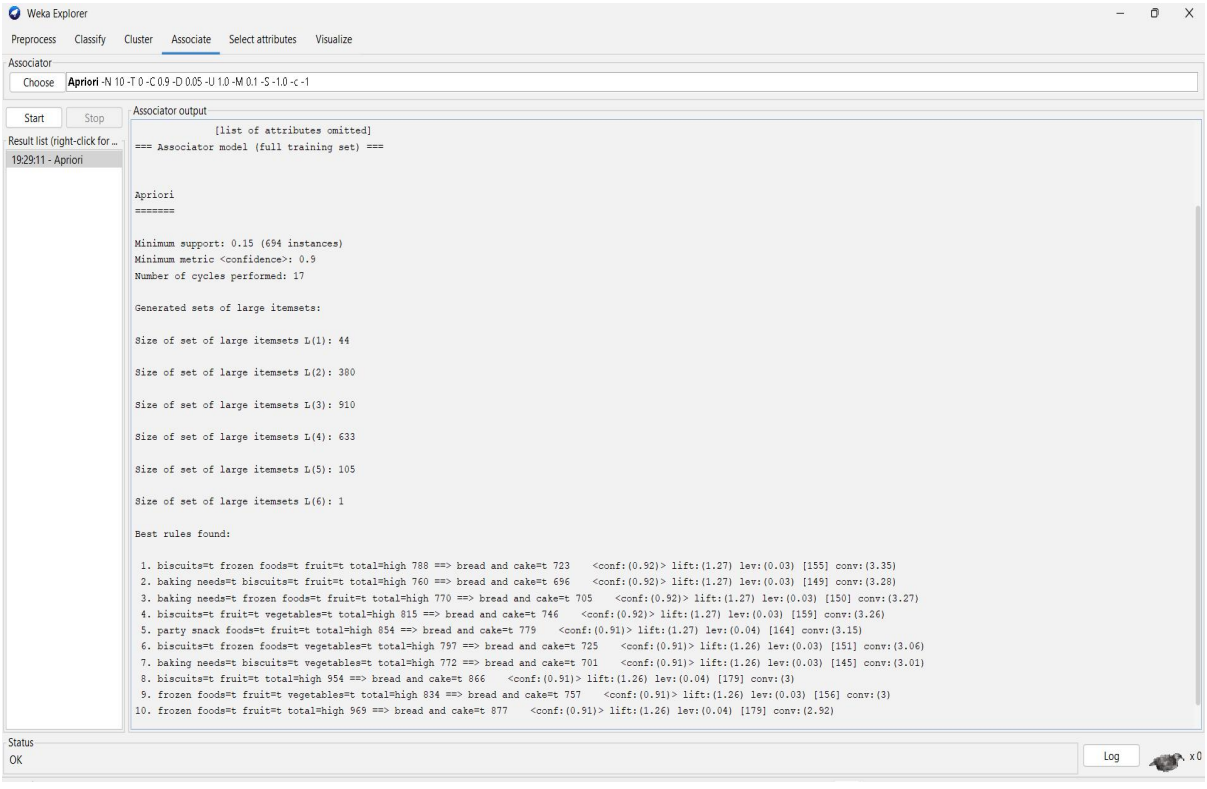




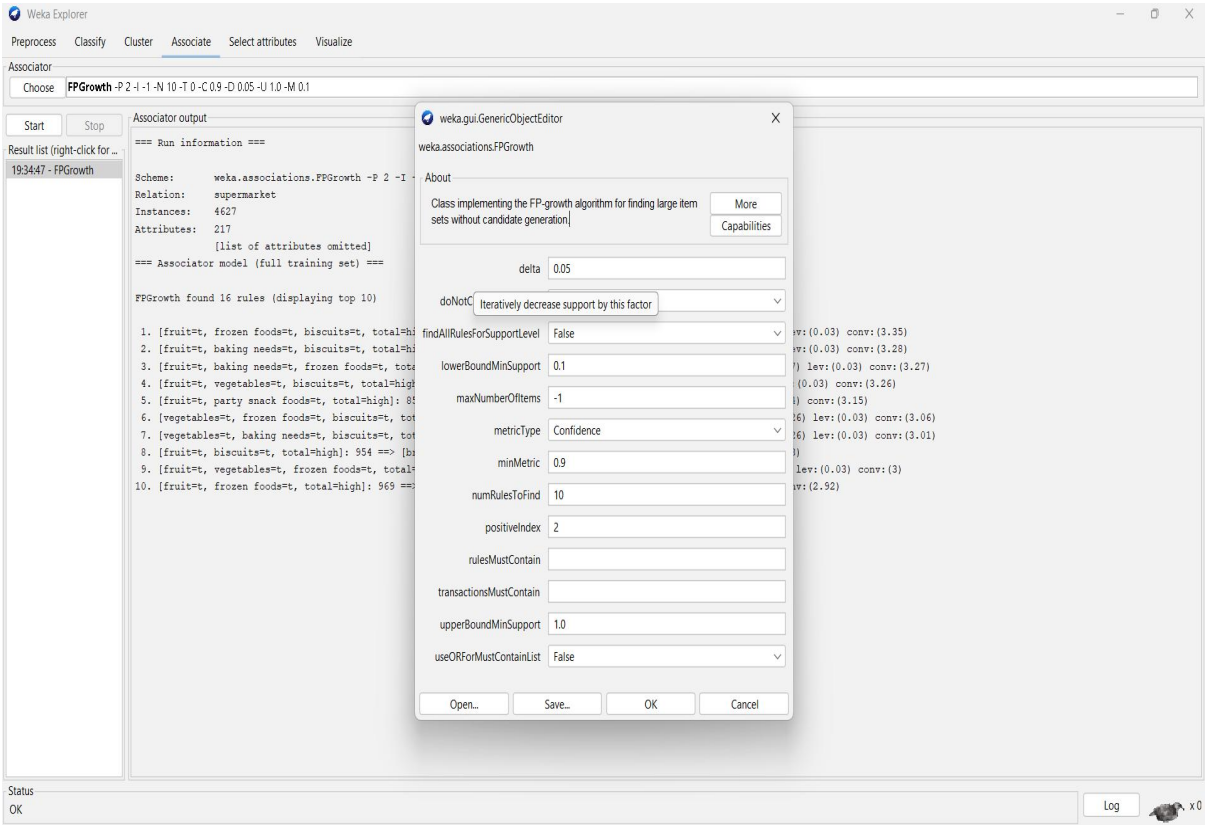


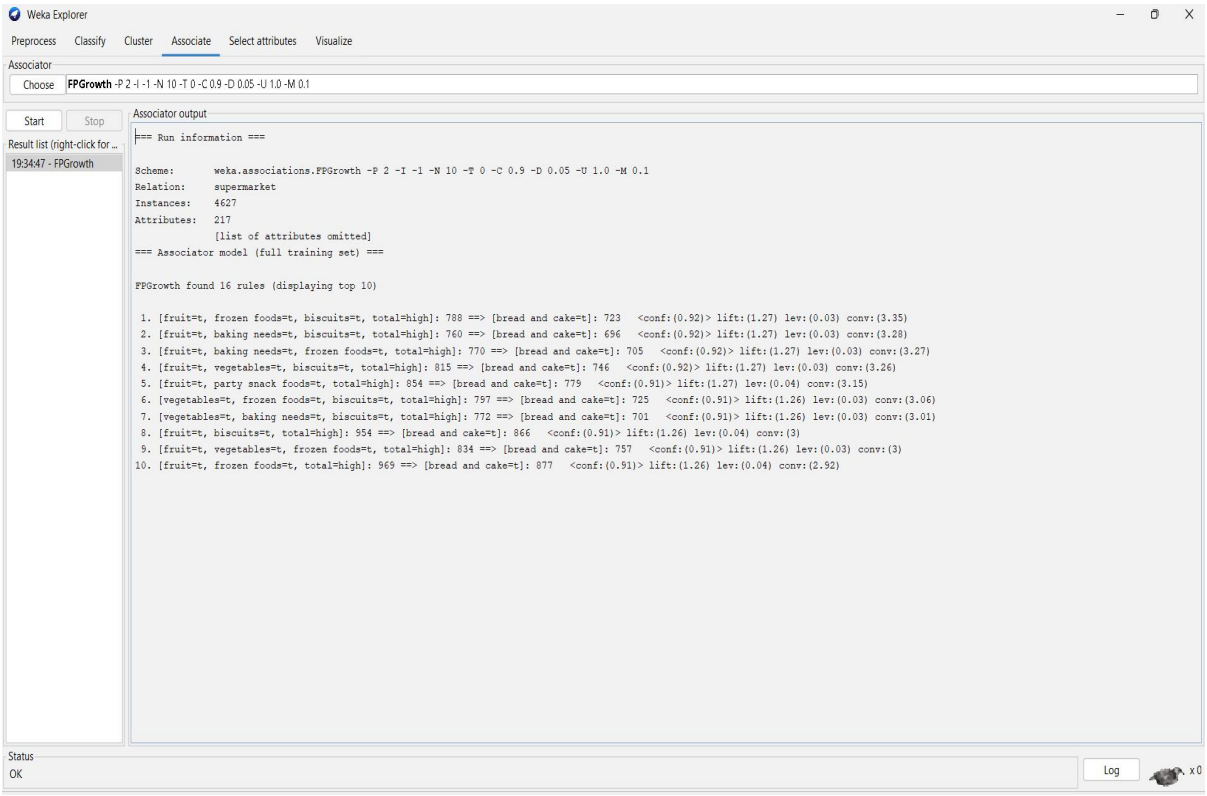
EXPERIMENT 25 : FREQUENT PATTERN MINING USING ASSOCIATION RULE THROUGH WEKA AND R -TOOLS



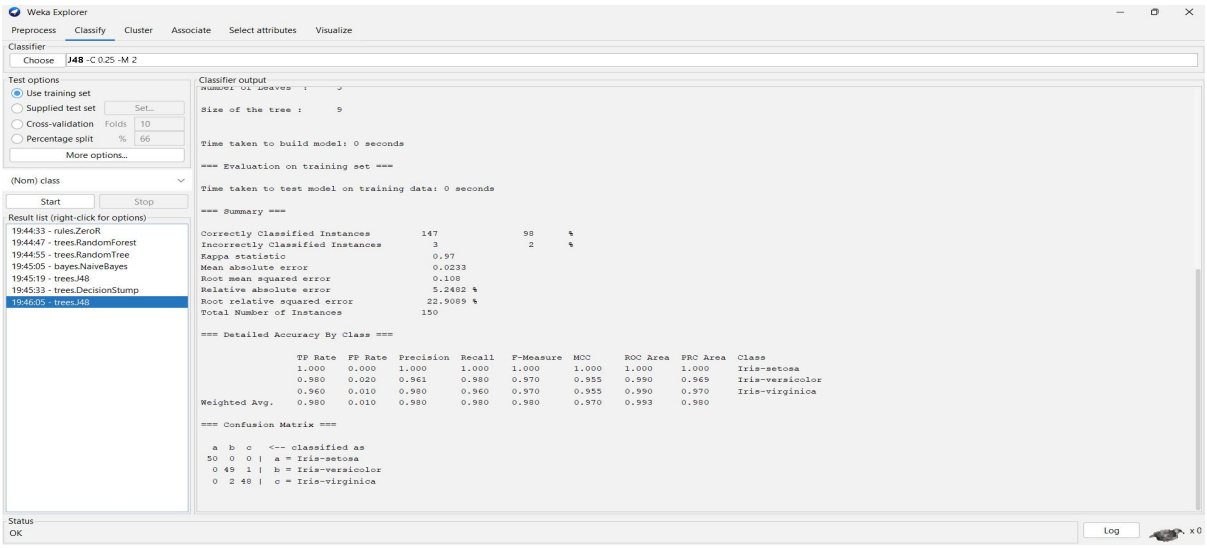


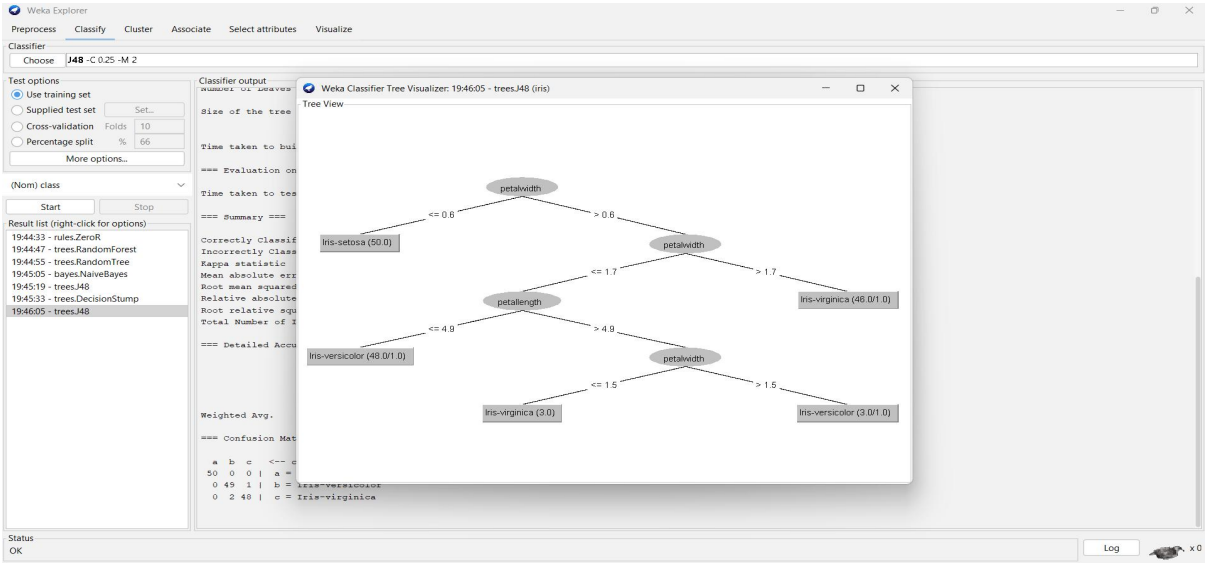
EXPERIMENT 26 : FFREQUENT PATTERN MINING USING FP GROWTH THROUGH WEKA TOOL

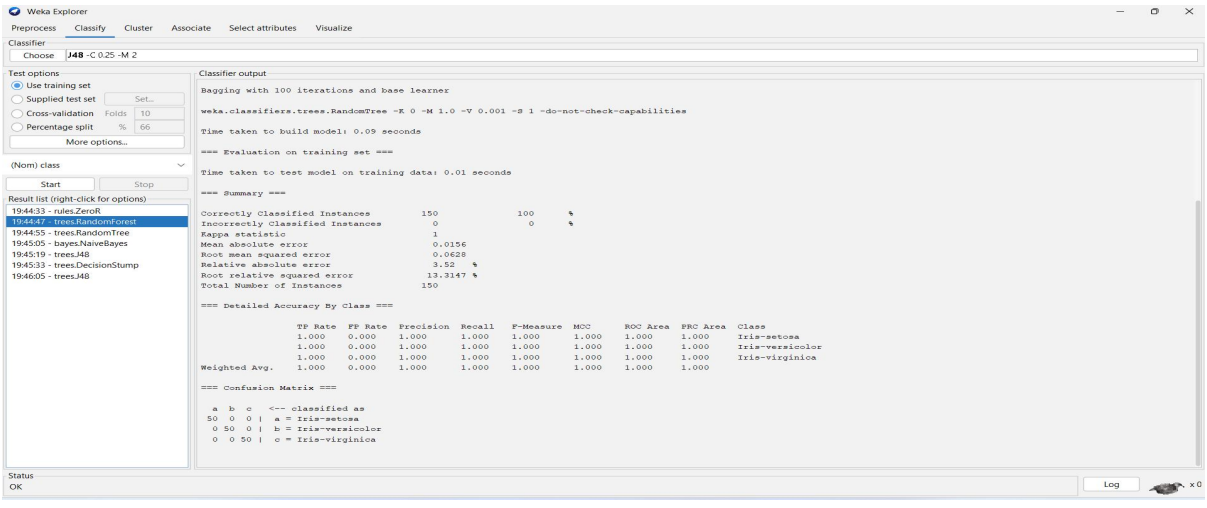


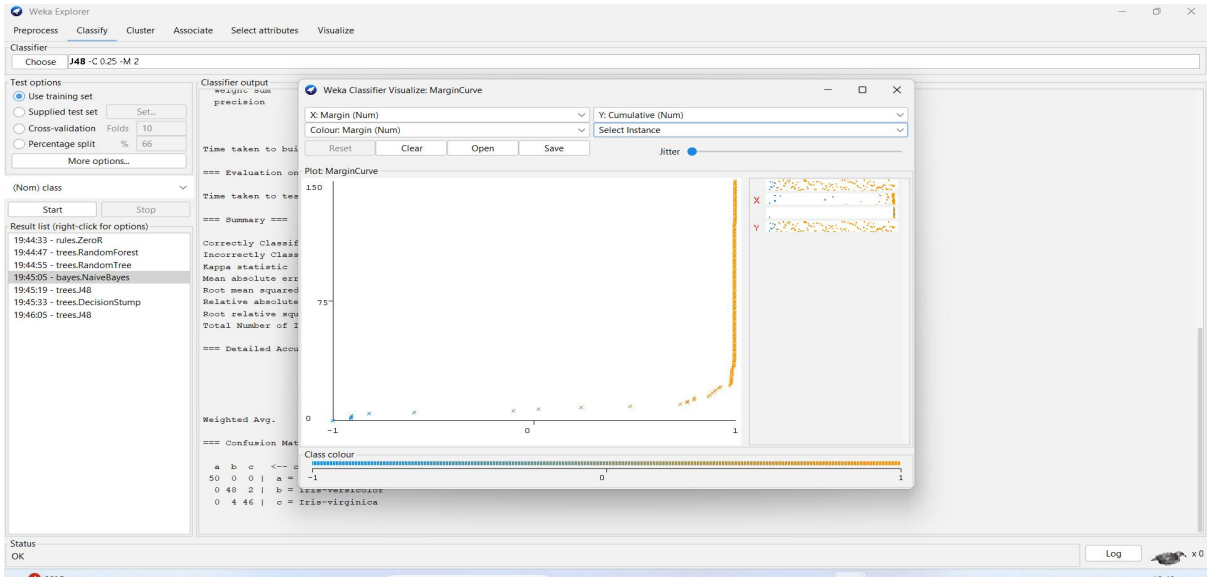


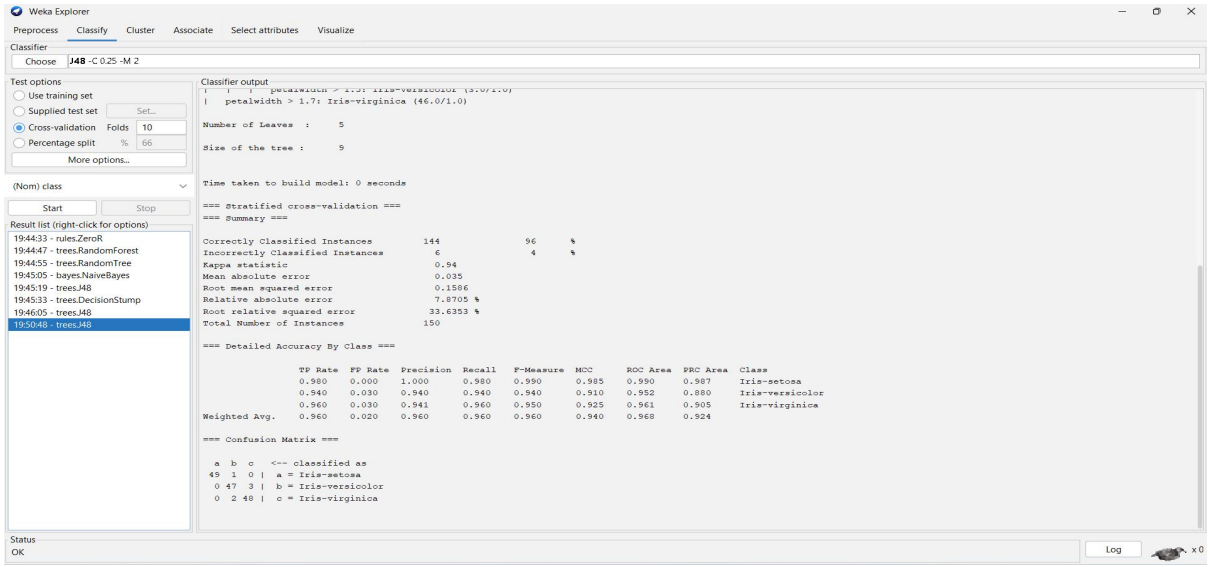
EXPERIMENT 27 : PREDICTION OF CATEGORICAL DATA USING DECISION TREE ALGORTIHM THROUGH WEKA

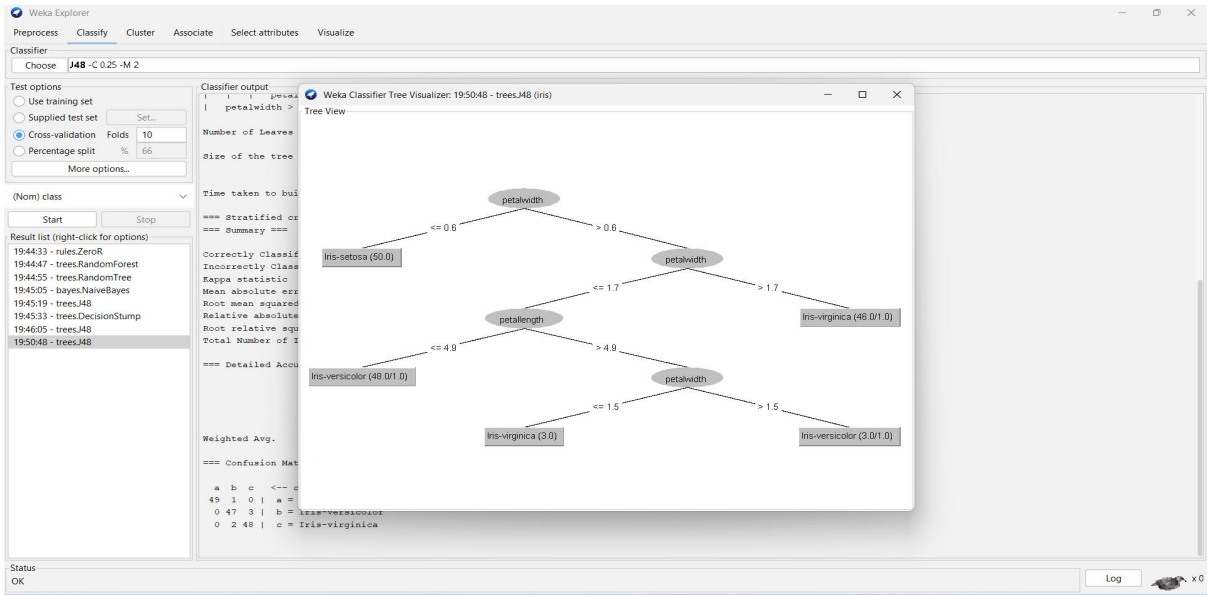




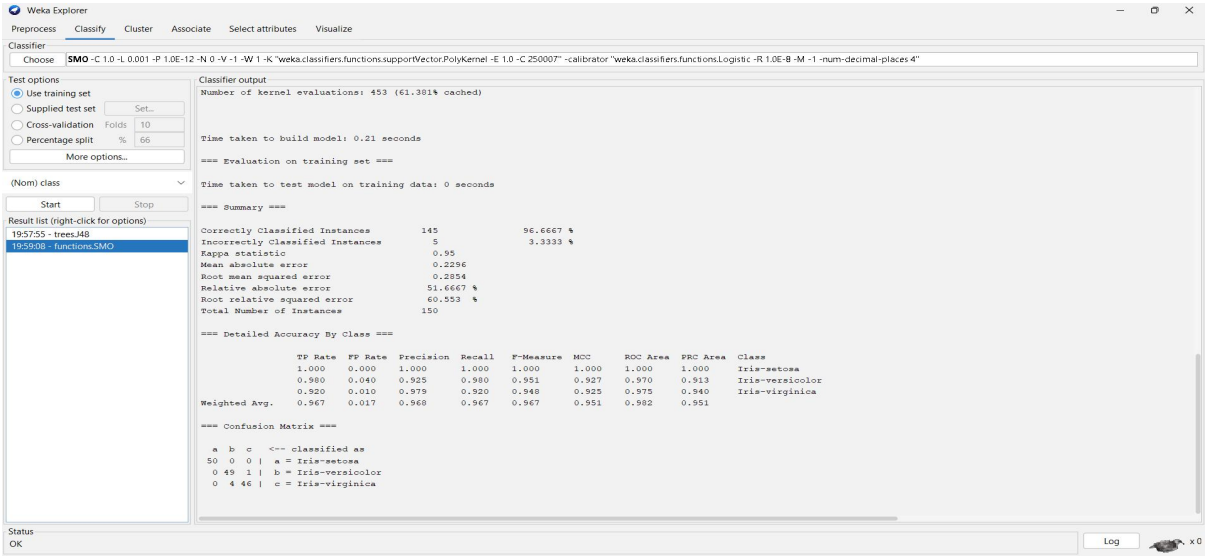


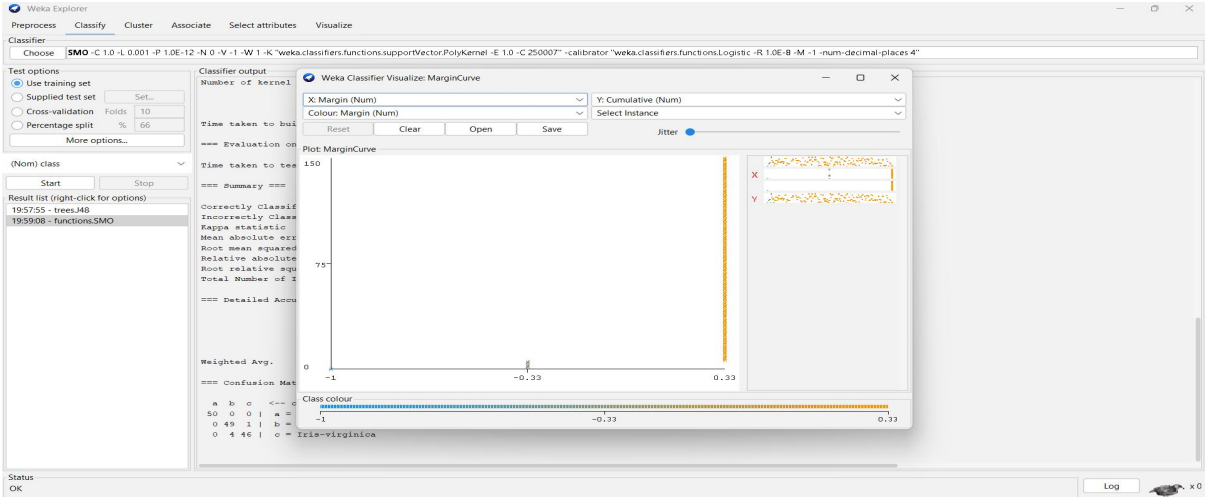

Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Classifier
Choose
J48 -C 0.25 -M 2
Test options
Use training set
Supplied test set
Set...
Cross-validation
Folds
10
Percentage split
%
66
More options...
(Nom) class
Start
Stop
Result list (right-click for options)
19:44:33 - rules.ZeroR
19:44:47 - trees.RandomForest
19:44:55 - trees.RandomTree
19:45:05 - bayes.NaiveBayes
19:45:19 - trees.J48
19:45:33 - trees.DecisionStump
19:46:05 - trees.J48
Classifier output
Weka Classifier Visualize: MarginCurve
X: Margin (Num)
Y: Cumulative (Num)
Colour: Margin (Num)
Select Instance
Reset
Clear
Open
Save
Jitter
Plot: MarginCurve
150
75
0
-1
0
1
Class colour
Status
OK
Log
x 0

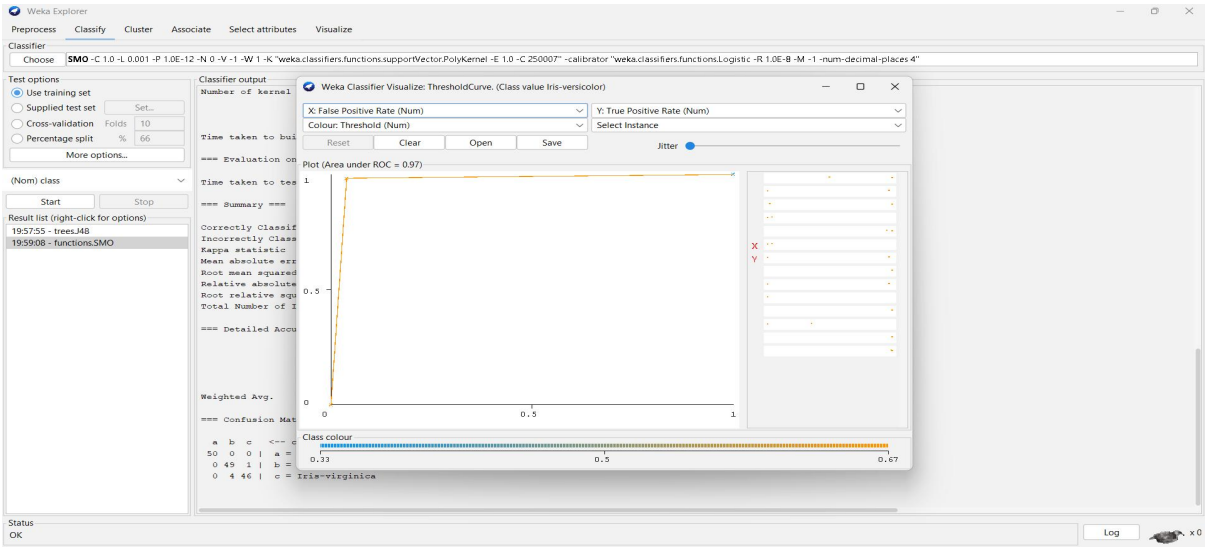

Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Classifier
Choose
J48 -C 0.25 -M 2
Test options
Use training set
Supplied test set
Set...
Cross-validation
Folds
10
Percentage split
%
66
More options...
(Nom) class
Start
Stop
Result list (right-click for options)
19:44:33 - rules.ZeroR
19:44:47 - trees.RandomForest
19:44:55 - trees.RandomTree
19:45:05 - bayes.NaiveBayes
19:45:19 - trees.J48
19:45:33 - trees.DecisionStump
19:46:05 - trees.J48
19:50:48 - trees.J48
Classifier output
| petalwidth > 1.7: Iris-virginica (46.0/1.0)
Number of Leaves : 5
Size of the tree : 9
Time taken to build model: 0 seconds
=== Stratified cross-validation ===
=== Summary ===
Correctly Classified Instances 144 96 %
Incorrectly Classified Instances 6 4 %
Kappa statistic 0.94
Mean absolute error 0.035
Root mean squared error 0.1586
Relative absolute error 7.8705 %
Root relative squared error 33.6353 %
Total Number of Instances 150
=== Detailed Accuracy By Class ===
TP Rate FP Rate Precision Recall F-Measure MCC ROC Area PRC Area Class
0.980 0.000 1.000 0.980 0.990 0.985 0.990 0.987 Iris-setosa
0.940 0.030 0.940 0.940 0.940 0.910 0.952 0.880 Iris-versicolor
0.960 0.030 0.941 0.960 0.950 0.925 0.961 0.905 Iris-virginica
Weighted Avg. 0.960 0.020 0.960 0.960 0.960 0.940 0.968 0.924
=== Confusion Matrix ===
a b c <-- classified as
49 1 0 | a = Iris-setosa
0 47 3 | b = Iris-versicolor
0 2 48 | c = Iris-virginica
Status
OK
Log
x 0


Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Classifier
Choose
J48 -C 0.25 -M 2
Test options
Use training set
Supplied test set
Set...
Cross-validation
Folds
10
Percentage split
%
66
More options...
(Nom) class
Start
Stop
Result list (right-click for options)
19:44:33 - rules.ZeroR
19:44:47 - trees.RandomForest
19:44:55 - trees.RandomTree
19:45:05 - bayes.NaiveBayes
19:45:19 - trees.J48
19:45:33 - trees.DecisionStump
19:46:05 - trees.J48
19:50:48 - trees.J48
Classifier output
Weka Classifier Tree Visualizer: 19:50:48 - trees.J48 (iris)
Tree View
petalwidth
<= 0.6
> 0.6
Iris-setosa (50.0)
petalwidth
<= 1.7
> 1.7
petallength
Iris-virginica (46.0/1.0)
<= 4.9
> 4.9
Iris-versicolor (48.0/1.0)
petalwidth
<= 1.5
> 1.5
Iris-virginica (3.0)
Iris-versicolor (3.0/1.0)
Status
OK
Log
x 0

EXPERIMENT 28 : PREDICTION OF CATEGORICAL DATA USING SMO ALGORTIHM THROUGH WEKA







EXPERIMENT 29 : EVALUATING ACCURACY OF THE CLASSIFIERS

